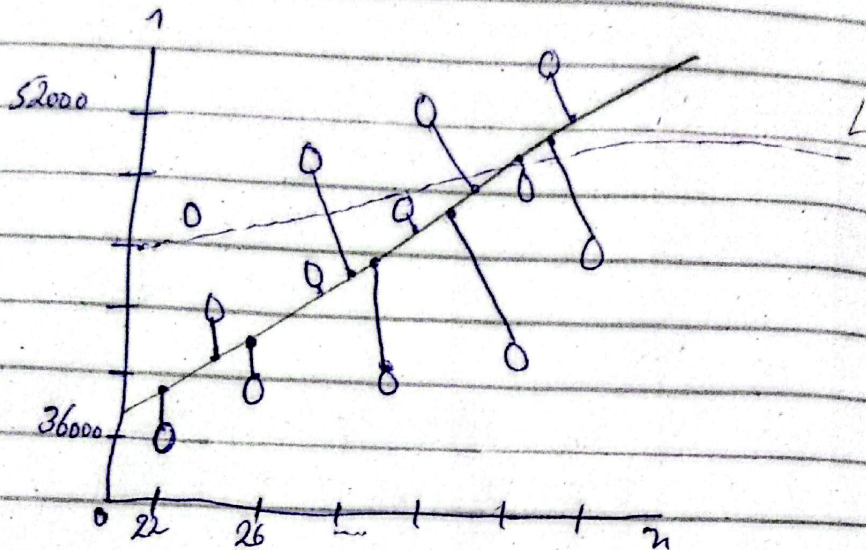


→ Residual error

Notes we create the scatter plot
on the basis of our main data points
like ex



so we have created this Now we will have
created best fit line.

so Residual error means, those data
point which are actual and the
data point which have predicted,
~~one~~ the difference between on both
is known as Residual error.

so Residual error means the difference
blw your actual data and predicted
data points.

so let's assume we have done sum
for all errors and after we notice
that one one point we get much error.
and then we will create those new line

so what we will do? we will create
multiple lines and when we draw a line
where there will be less error for other
lines then that line will be selected.

so for that we will use the
Mean Squared Error (MSE)

MSE: is one of the
most fundamental metrics used
in statistics, data science, and ML.
it measures the average squared
difference b/w actual values (observed data)
and predicted values (model output)

What is Mean Squared Error
used for?

MSE serves two primary purposes:
1. Model Evaluation and Performance Metric

Quantifies Model Accuracy: MSE gives
you a single number that tells
you how close your model's predictions
are to the actual true values

- Comparing Different Models:
You can compare two or more regression models on the same dataset, the model with the lower MSE is considered to have better overall predictive accuracy.

2. Optimization/Loss Functions-

- Guiding Model Training: Algorithms like Linear Regression and Neural Networks use MSE as their loss function (or cost function).
- During training, the algorithm adjusts its parameters (like slope m and intercept b) continuously to minimize the MSE.

Mathematical Formula:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Where

- n = Total number of data points/objects
- y_i = Actual value for the i -th datapoint
- $\hat{y}_i$ = predicted value from your model ($y_{\text{predicted}}$)
- $(y_i - \hat{y}_i)$ = Error (Residual) for each individual point

ex: (Using previous Data)

x	y_i	$\hat{y}_i$	$y_i - \hat{y}_i$	$(y_i - \hat{y}_i)^2$
1	1.50	1.58	-0.08	0.0064
2	3.80	3.99	-0.19	0.0361
3	6.70	6.40	+0.30	0.0900
4	9.00	8.81	+0.19	0.0361
5	11.20	11.22	-0.02	0.0004
6	13.60	13.63	-0.03	0.0009
7	16.00	16.04	-0.04	0.0016

calculations

Sum of Squared Errors

$$= 0.0064 + 0.0361 + 0.0900 + 0.0361 + 0.0004 + 0.0009 + 0.0016 = \boxed{0.1715}$$

$$MSE = \frac{0.1715}{7} = \boxed{0.0245}$$

interpretation: An MSE of 0.0245 is very close to 0, which confirms that your line of best fit $y = 2.41x - 0.83$ fits your data exceptionally well!

Cost functions it states that whatever residual errors comes at the end add them (do sum of all)

Note: MSE is a specific formula that measures average squared errors whereas a cost function is a general machine learning concept representing any function to reduce errors

In short: MSE is a specific type of cost function commonly used for linear regression. (just like cross-Entropy is a cost function used for classification).

For:

1/ Goal of ML: Minimize the cost function, $J(m,b)$

2/ MSE is the standard cost function for Linear Reg.

3/ Gradient Descent is the engine that steps down the cost "bowl" to find ideal m and b .

Cost function vs loss function

While people often use these terms interchangeably, there is a technical distinction.

↳ loss function:

Measures the error for a single data point:

$$\text{Loss} = (y_i - \hat{y}_i)^2$$

↳ Cost function: Measures the average error across the entire dataset (n points)

$$J(m, b) = \frac{1}{2n} \sum_{i=1}^n (m x_i + b - y_i)^2$$